

READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

The History of the Pencil

The history of the pencil starts with a violent thunderstorm. When some particularly ferocious weather struck the Lake District in North West England in the sixteenth century, locals in the village of Borrowdale discovered a large uprooted tree. Underneath the tree lay an unknown black substance which we now know as graphite, was slightly shiny and smooth to the touch. And it left a black smear on the hands of all who touched it.

Initially, the local farmers used the newly discovered material as a handy way to identify their sheep. However, others quickly realised the potential for using this intriguing substance to write on paper. When it was untreated the material was very soft, which meant that it was messy to handle. To make it fit for use with paper, people enclosed a thin core of the substance in stiff sheep hides or rope. At this time chemistry was still in its infancy. People searched for a word to describe this increasingly useful substance and came up with plumbago which, in Latin, means acts or writes like lead. Later the name was changed to graphite. But because words have remarkable staying power, we still call graphite the lead of a pencil even though it is now known that there is no trace of real lead in graphite.

Graphite has a very high melting point at around 3,500 degrees Celsius. This made it invaluable to the British army and navy as a secret ingredient in the manufacture of cannon balls. The Royal Ordnance, or weaponry section of the British armed forces, used graphite as a lining inside the moulds for cannon balls, which, as a result, the British could turn out faster and more cheaply than their European rivals. In addition to its value to the armed forces, the government quickly realised the commercial potential of the graphite at Borrowdale, and assumed control of all the mines there during the sixteenth century. Armed guards accompanied the precious graphite all the way down to the metal foundries by the naval shipyards in the south of England. The graphite was so valuable that the locals, who called it wad, started to steal it. As a deterrent, an act of Parliament in 1752 made this offence punishable by time in prison.

The Italians originally invented the wooden casing to hold a thin rod of plumbago firmly in place for ease of writing. Italian craftsmen hollowed out two small sections of cedar wood, into one of which they laid the lead. They then glued the other section over the top and left the two halves to set. When dry, the whole apparatus formed what today we know as a pencil. The Germans took this technique and developed it further by applying mass-production techniques to pencils. At the same time Nicolas-Jacques Conté, a French officer in Napoleon Bonaparte's army during the late 1700s, developed a method of mixing powdered graphite and clay together for firing in a kiln. Adding more clay to the mixture

helped make the pencil harder, sharper, and more precise in its mark. More graphite helped make a pencil mark that was softer, thicker and darker.

The varying quality of pencil leads eventually gave rise to a system for categorizing the fineness of the pencil mark. Pencil manufacturers all over the world still use this so-called HB grading system today. The H stands for the Hardness of the pencil while the B stands for its Blackness. An HB pencil is a standard pencil and a variety of letters and numbers are used to designate different types of lead.

Significant seams of graphite exist in parts of China, which now produces most of the world's pencils. Interestingly, the Borrowdale mine in the Lake District remains the only significant source of graphite in its near-pure form in the world. Nowadays the highest grade of graphite at Borrowdale is totally exhausted, although other grades can still be found, and England's pencil industry continues to thrive in the nearby town of Keswick.

The pencil has turned out to be a remarkably resilient and valuable tool whose use has survived well into our high-tech times, as a well-known story shows. It is sometimes said that the American space programme spent millions of dollars to invent a pen capable of writing in the zero gravity of space. The Russians, by contrast, simply equipped their astronauts with good old-fashioned pencils that never let them down. It should be pointed out though, that the popular myth about Americans overlooking the practical advantages of pencils in zero gravity is merely fiction. In actual fact, both American and Russian astronauts were equipped with pencils in their respective countries' first space flights. A private company later developed pens for writing in zero gravity. In fact, astronauts of every nation now use pens. But no matter — pencils remain in use in every classroom, every planning, building and drawing office, and in every art studio in the world. And there is nothing to suggest that we are likely to invent anything better than graphite to use in our pencils.

Questions 1–6

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 1–6 on your answer sheet.

The early history of graphite in Britain

- Graphite was first found below a **1** _____ blown down in a storm.
- The first use of graphite was to make marks on **2** _____.
- The characteristics of raw graphite:
 - dirty to use because it is so **3** _____.
 - originally wrapped in **4** _____ or animal skin to make it useable as a pencil.
- Graphite came to the notice of the government for military and commercial purposes.
 - The government completely took over the **5** _____ at Borrowdale.
 - They employed guards to protect the graphite on its journey south.
- Local people began to **6** _____ graphite for the money involved.
 - The government passed a law to protect the graphite industry.

Questions 7–13

Do the following statements agree with the information given in Reading Passage 1?

In boxes 7–13 on your answer sheet, write

TRUE	<i>if the statement agrees with the information</i>
FALSE	<i>if the statement contradicts the information</i>
NOT GIVEN	<i>if there is no information on this</i>

- 7 The Italians were the first to make pencils out of wood.
- 8 The Germans used different types of wood to produce pencils.
- 9 More clay in a pencil makes it write more darkly.
- 10 After the HB code was introduced, it very quickly became used by all manufacturers.
- 11 English pencil factories have now all closed down.
- 12 American astronauts used pencils on their early journeys into space.
- 13 The use of graphite pencils is unlikely to continue into the future.

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Questions 1–6 填空题

题号	答案	题干翻译	详细定位与原文翻译	详细解释
1	tree	石墨最早是在一棵被暴风雨吹倒的 _____ 下面被发现的。	第1段: <i>locals in the village of Borrowdale discovered a large uprooted tree. Underneath the tree lay an unknown black substance which we now know as graphite.</i> 译: Borrowdale 村的当地人发现了一棵被连根拔起的大树。在这棵树下面, 躺着一一种未知的黑色物质, 也就是我们现在所知道的石墨。	题干中的 first found below 对应原文 Underneath the tree lay ; blown down in a storm 对应 violent thunderstorm / uprooted tree 。空格前有 a , 所以填单数名词 tree 。
2	sheep	石墨最早的用途是在 _____ 身上做标记。	第2段: <i>Initially, the local farmers used the newly discovered material as a handy way to identify their sheep.</i> 译: 起初, 当地农民把这种新发现的材料当作一种方便的方法, 用来识别他们的羊。	题干中的 first use 对应原文 Initially ; make marks on 是对 identify their sheep 的改写。农民用石墨给羊做记号, 以便辨认, 所以答案是 sheep 。
3	soft	原始石墨的特点: 因为它太 _____, 所以使用起来很脏。	第2段: <i>When it was untreated the material was very soft, which meant that it was messy to handle.</i> 译: 未经处理时, 这种材料非常柔软, 这意味着它拿起来很脏、很容易弄脏手。	题干中的 dirty to use 对应原文 messy to handle ; because it is so 对应原文原因 very soft 。所以填 soft 。
4	rope	原本为了让它可以作为铅笔使用, 人们把它包在 _____ 或动物皮里。	第2段: <i>people enclosed a thin core of the substance in stiff sheep hides or rope.</i> 译: 人们把这种物质细细的一芯包在硬羊皮或绳子里。	题干中的 wrapped in 对应原文 enclosed ... in ; animal skin 对应 stiff sheep hides 。原文结构是 sheep hides or rope , 题干变成 rope or animal skin , 所以空格填 rope 。
5	mines	政府完全接管了 Borrowdale 的 _____。	第3段: <i>the government quickly realised the commercial potential of the graphite at Borrowdale, and assumed control of all the mines there during the sixteenth century.</i> 译: 政府很快意识到 Borrowdale 石墨的商业潜力, 并在16世纪接管了那里的所有矿山。	题干中的 completely took over 对应原文 assumed control of all ; at Borrowdale 对应 there 。注意答案是复数 mines , 因为原文是 all the mines 。
6	steal	当地人因为其中涉及的钱财, 开始 _____ 石墨。	第3段: <i>The graphite was so valuable that the locals, who called it wadd, started to steal it.</i> 译: 石墨非常值钱, 以至于当地人——他们称它为 wadd——开始偷石墨。	题干中的 for the money involved 对应原文 so valuable ; began to 对应 started to 。所以答案是动词原形 steal 。

Questions 7–13 判断题 TRUE / FALSE / NOT GIVEN

题号	答案	题干翻译	详细定位与原文翻译	详细解释
7	TRUE	意大利人是最早用木头制作铅笔的人。	第4段: <i>The Italians originally invented the wooden casing to hold a thin rod of plumbago firmly in place for ease of writing.</i> 译: 意大利人最初发明了木质外壳, 用来牢牢固定一根细的石墨棒, 以方便书写。	题干说 the first to make pencils out of wood , 原文说 originally invented the wooden casing 。虽然原文没有直接说“first”, 但 originally invented 就表示最早发明, 所以判断为 TRUE 。
8	NOT GIVEN	德国人使用不同种类的木材来生产铅笔。	第4段: <i>The Germans took this technique and developed it further by applying mass-production techniques to pencils.</i> 译: 德国人采用了这项技术, 并通过将大规模生产技术应用到铅笔制造来进一步发展它。	原文只说德国人发展了量产技术, 没有说他们是否使用了不同种类的木材。题干中的 different types of wood 完全没有对应信息, 所以是 NOT GIVEN 。
9	FALSE	铅笔中加入更多黏土会让它写出来更黑。	第4段: <i>Adding more clay to the mixture helped make the pencil harder, sharper, and more precise in its mark. More graphite helped make a pencil mark that was softer, thicker and darker.</i> 译: 加入更多黏土会让铅笔更硬、更尖, 笔迹更精确。加入更多石墨则会让笔迹更软、更粗、更黑。	题干说 more clay → more darkly , 但原文说 more graphite → darker , 而 more clay → harder, sharper, more precise 。题干把 clay 和 graphite 的作用混淆了, 所以是 FALSE 。
10	NOT GIVEN	HB 编码引入之后, 很快被所有制造商使用。	第5段: <i>Pencil manufacturers all over the world still use this so-called HB grading system today.</i> 译: 世界各地的铅笔制造商今天仍然使用这个所谓的 HB 分级系统。	原文只说现在全世界制造商仍在使用 HB 系统, 但没有说明它被引入之后是否 very quickly 被所有制造商采用。题干考的是“速度”, 原文没有这个信息, 所以是 NOT GIVEN 。
11	FALSE	英国的铅笔工厂现在都已经关闭了。	第6段: <i>England's pencil industry continues to thrive in the nearby town of Keswick.</i> 译: 英格兰的铅笔产业在附近的 Keswick 镇仍然继续繁荣发展。	题干说英国铅笔工厂 all closed down , 而原文说英国铅笔产业 continues to thrive , 即仍然繁荣。两者直接矛盾, 所以是 FALSE 。
12	TRUE	美国宇航员在早期太空旅行中使用过铅笔。	第7段: <i>In actual fact, both American and Russian astronauts were equipped with pencils in their respective countries' first space flights.</i> 译: 事实上, 美国和俄罗斯宇航员在各自国家最早的太空飞行中都配备了铅笔。	题干中的 early journeys into space 对应原文 first space flights ; used pencils 对应 were equipped with pencils 。所以是 TRUE 。
13	FALSE	石墨铅笔的使用不太可能延续到未来。	第7段: <i>pencils remain in use in every classroom, every planning, building and drawing office, and in every art studio in the world. And there is nothing to suggest that we are likely to invent anything better than graphite to use in our pencils.</i> 译: 铅笔仍然被用于世界上每一间教室、每一个规划、建筑和绘图办公室, 以及每一个艺术工作室。而且没有任何迹象表明, 我们可能会发明出比石墨更适合用于铅笔的东西。	题干说 unlikely to continue into the future , 但原文强调铅笔仍广泛使用, 并且没有迹象表明会出现比石墨更好的替代物。题干与原文方向相反, 所以是 FALSE 。